



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN
PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (*INFORMATION SHEET*)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-2: 2015CORE ABILITIES	
TAHAP / LEVEL	2 (TWO)	
NO. DAN TAJUK KEBOLEHAN TERAS <i>ICORE ABILITY NO. AND TITLE</i>	CA04 HEALTH, SAFETY AND ENVIRONMENTAL AWARENESS	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	04.01 DEMONSTRATE HEALTH COMPLIANCE 04.02 DEMONSTRATE SAFETY COMPLIANCE 04.03 DEMONSTRATE ENVIRONMENT COMPLIANCE	
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TAJUK/TITLE: **ENVIRONMENTAL ACT, RULES AND REGULATION**

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on the ability to demonstrate level of understanding in Health, Safety and Environment adaptation and apply it to work activities. The person who is competent in this CA shall be able to adapt to HSE by executing his/her duties while observing Health, Safety and Environment Rules and Regulations.

PENERANGAN/INFORMATION:

1. Environmental Requirements

In the promotion of environmentally sound and sustainable development, the Government of Malaysia has established the necessary legal and institutional arrangements such

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that environmental factors are considered at the early stages of project planning. With reference to the licensing requirements for establishment of business/industry in the country, environmental requirements and assessment constitute the second level of approval that need to be obtained after a business or industry has been registered.

1.1 Environmental Quality Act, 1974

The legislation that is related to the prevention, abatement, control of pollution and enhancement of the environment in Malaysia is the Environmental Quality Act, 1974. The Act restricts the discharge of wastes into the environment in contravention of the acceptable conditions. To date 38 sets of Regulations and Orders have been introduced and enforced. The Director General of Environmental Quality has been appointed by the Minister to administer this Act and any regulations and orders made thereunder.

Under the Environmental Quality Act (EQA), 1974 and the Regulations thereunder, industrial activities are required to obtain the following approvals from the Director General of Environmental Quality prior to project implementation:

- i) Environmental Impact Assessment (EIA) reports-under Section 34A of the EQA, 1974 (for prescribed activities);
- ii) Site suitability evaluation (for non-prescribed activities);
- iii) Written permission to construct-under Section 19 of the EQA, 1974 (for prescribed premises (scheduled wastes treatment and disposal facilities, crude palm oil mills and raw natural rubber processing mills);
- iv) The owner must submit a written declaration to the Director General who confirmed that the design and construction of pollution control systems are in compliance with specifications . Environmental Quality (Clean Air) Regulations 2014.
- v) License to use and occupy prescribed premises and prescribed conveyances - under Section 18 of the EQA, 1974.

Environmental laws and regulations one of the three strategies embodied in EQA, 1974, is for the regulation of pollution. The other two strategies are for preventing and abating any form of pollution. To bring the law and other environmentally-related laws into effect, the laws and regulations listed below have been introduced and are strictly enforced by the Department of Environment.

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- i) Control of agro-based water pollution
 - a) Environmental Quality (Licensing) Regulations, 1977
 - b) Environmental Quality (Prescribed Premises) (Crude Palm Oil) Order, 1977
 - c) Environmental Quality (Prescribed Premises) (Crude Palm Oil) Regulations, 1977, and (Amendment) 1982
 - d) Environmental Quality (Prescribed Premises) (Raw Natural Rubber) Order, 1978
 - e) Environmental Quality (Prescribed Premises) (Raw Natural Rubber) Regulations, 1978
- ii) Control of municipal and industrial waste water pollution
 - a) Environmental Quality (Industrial Effluent) Regulations, 2009
 - b) Environmental Quality (Sewage) Regulations, 2009
 - c) Environmental Quality (Control of Pollution From Solid Waste Transfer Station and Landfill) Regulations, 2009
 - d) Environmental Quality (Prohibition on the Use of Controlled Substance in Soap, Synthetic Detergent and Other Cleaning Agents) Order, 1995
- iii) Control of industrial emissions
 - a) Environmental Quality (Clean Air) Regulations, 2014
 - b) Penalty / Enforcement Action
- iv) Control of motor vehicle emissions
 - a) Environmental Quality (Control of Emission From Diesel Engines) Regulations, 1996
 - b) Environmental Quality (Control of Emission From Petrol Engines) Regulations, 1996
 - c) Environmental Quality (Control of Lead Concentration in Motor Gasoline) Regulations, 1985
 - d) Environmental Quality (Motor Vehicle Noise) Regulations, 1987
- v) Control of toxic and hazardous waste management

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- a) Environmental Quality (Scheduled Wastes) Regulations, 2005
- b) Environmental Quality (Prescribed Premises) (Scheduled Wastes Treatment and Disposal Facilities) Regulations, 1989 [P.U.(A) 141/89]
- c) Environmental Quality (Prescribed Premises) (Scheduled Wastes Treatment and Disposal Facilities) Order, 1989
- d) Environmental Quality (Prescribed Conveyance)(Scheduled Wastes) Order 2005 - P.U.(A) 293/2005
- e) Promotion of Investments (Promoted Activities and Products) (Amendment) (No. 10) Order, 1990 (made under the Promotion of Investments Act, 1986)
- vi) Open Burning Prevention
 - a) Prohibition of open burning under Section 29A, EQA 1974
 - b) Environmental Quality (Declared Activities)(Open Burning) Order, 2003

1.2 Environmental Impact Assessment (EIA) For Prescribed Activities

All prescribed activities need to obtain EIA approval from the Director General of Environment prior to the giving of approval by the relevant Federal or State Government authority for the implementation of the project. The Approving Authority is the Government Authority that has the task of deciding, whether or not a project should proceed. List of prescribed activities are:

- i) Agriculture
 - a) Land development schemes covering an area of 500 hectares or more to bring forest land into agriculture production.
 - b) Agriculture programmes necessitating the resettlement of 100 families or more.
 - c) Development of agricultural estates covering an area of 500 hectares or more involving changes in type of agricultural use.
- ii) Airport
 - a) Construction of airports (having an airstrip of 2,500 meters or longer).
 - b) Airstrip development in state and national parks.
- iii) Drainage And Irrigation

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- a) Construction of dams and man-made lakes and artificial enlargement of lakes with surface areas of 200 hectares or more.
 - b) Drainage of wetland, wild-life habitat or of virgin forest covering an area of 100 hectares or more.
 - c) Irrigation schemes covering an area of 5,000 hectares or more.
- iv) Land Reclamation
 - a) Coastal reclamation involving an area of 50 hectares or more.
- v) Fisheries
 - a) Construction of fishing harbours.
 - b) Harbour expansion involving an increase of 50 per cent or more in fish landing capacity per annum.
 - c) Land based aquaculture projects accompanied by clearing of mangrove swamp forests covering an area of 50 hectares or more.
- vi) Forestry
 - a) Conversion of hill forest land to other land use covering an area of 50 hectares or more.
 - b) Logging or conversion of forest land to other land use within the catchments area of reservoirs used for municipal water supply, irrigation or hydropower generation or in areas adjacent to state and national parks and national marine parks.
 - c) Logging covering an area of 500 hectares or more.
 - d) Conversion of mangrove swamps for industrial, housing or agricultural use covering an area of 50 hectares or more.
 - e) Clearing of mangrove swamps in islands adjacent to national marine parks.
- vii) Housing
 - a) Housing development covering an area of 50 hectares or more.
- viii) Industry
 - a) Chemical –Where production capacity of each product or of combined products is greater than 100 tonnes/ day.
 - b) Petrochemicals -All sizes.

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c) Non-ferrous -Primary smelting:

- Aluminium -all sizes
- Copper -all sizes
- Others -producing 50 tonnes/day and above of product.

d) Non -Metallic -Cement-for clinker through put of 30 tonnes/hour and above.

-Lime -100 tonnes/day and above burnt lime rotary kiln or 50tonnes/day and above vertical kiln.

e) iron and Steel - Require iron ore as raw materials for production greater than 100tonnes/day; or using scrap iron as raw materials for production greater than 200 tonnes/day

f) Shipyards - Dead Weight Tonnage greater than 5000 tonnes.

g) Pulp and Paper- Production capacity greater than 50 tonnes/day.
Industry

ix) Infrastructure

a) Construction of hospitals without fall into beachfronts used for recreational purposes.

b) Industrial estate development for medium and heavy industries covering an area of 50 hectares or more.

c) Construction of expressways.

d) Construction of national highways.

e) Construction of new townships.

x) Ports

a) Construction of ports

b) Port expansion involving an increase of 50 per cent or more in handling capacity per annum.

xi) Mining

a) Mining of minerals in new areas where the mining lease covers a total area in excess of 250 hectares.

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- b) Ore processing, including concentrating for aluminium, copper, gold or tantalum.
- c) Sand dredging involving an area of 50 hectares or more.

xii) Petroleum

- a) Oil and gas fields development.
- b) Construction of off-shore and on-shore pipelines in excess of 50 kilometers in length.
- c) Construction of oil and gas separation, processing, handling, and storage facilities.
- d) Construction of oil refineries.
- e) Construction of product depots for the storage of petrol, gas or diesel (excluding service stations) which are located within 3 kilometres of any commercial, industrial or residential areas and which have a combined storage capacity of 60,000 barrels or more.

xiii) Power Generation And Transmission

- a) Construction of steam generated power stations burning fossil fuels and having a capacity of more than 10 megawatts.
- b) Dams and hydroelectric power schemes with either or both of the following:
- c) dams over 15 meters high and ancillary structures covering a total area in excess of 40 hectares;
- d) Reservoirs with a surface area in excess of 400 hectares.
- e) Construction of combined cycle power stations.
- f) Construction of nuclear-fueled power stations.

xiv) Quarries

Proposed quarrying of aggregate, limestone, silica quartzite, sandstone, marble and decorative building stone within 3 kilometres of any existing residential, commercial or industrial areas, or any area for which a licence, permit or approval has been granted for residential, commercial or industrial development.

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xv) Railways

- a) Construction of new routes.
- b) Construction of branch lines.

xvi) Transportation

Construction of Mass Rapid Transport projects.

xvii) Resort And Recreational Development

- a) Construction of coastal resort facilities or hotels with more than 80 rooms.
- b) Hill station resort or hotel development covering an area of 50 hectares or more.
- c) Development of tourist or recreational facilities in national parks.
- d) Development of tourist or recreational facilities or islands in surrounding waters which are gazetted as national marine parks.

xviii) Waste Treatment And Disposal

a) Toxic and Hazardous Waste

- Construction of incineration plant
- Construction of recovery plant (off-site)
- Construction of wastewater treatment plant (off-site)
- Construction of secure landfill facility
- Construction of storage facility (off-site)

b) Municipal Solid Waste

- Construction of incineration plant
- Construction of composting plant
- Construction of recovery/recycling plant
- Construction of municipal solid waste landfill facility

c) Municipal Sewage

- Construction of wastewater treatment plant
- Construction of marine out fall.

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xix) Water Supply

- a) Construction of dams, impounding reservoirs with a surface area of 200 hectares or more.
- b) Groundwater development for industrial, agricultural or urban water supply of greater than 4,500 cubic meters per day.

1.3 Site Suitability Evaluation (For Non-Prescribed Activities)

Due to the sensitivity of the project and polluting potential (significant impacts) from the construction and/or operations, some of the prescribed activities have been required to go through the Detailed EIA Procedures which involves public participation. One of the most important procedures which have an immediate bearing on the period of processing and condition of approval on environmental ground is the SITE SUITABILITY for the proposed project. Irrespective of whether the proposed industrial activity is going to be sited within an industrial estate or otherwise, it should be developed and managed with environmentally sound control measures. Therefore, all potential industrial sites for the establishment of new industrial activities which are NOT subject to EIA Order, 1987, particularly the Small and Medium Scale Industries (SMLs), are advised to refer to the Department of Environment for consideration and advice on site suitability. Those activities are:

- i) Iron and steel Industry.
- ii) Pulp and paper mills.
- iii) Cement plant.
- iv) Construction of coal fired power plant.
- v) Construction of dams and hydroelectric power schemes.
- vi) Land reclamation.
- vii) Incineration plant (scheduled waste & solid waste).
- viii) Sanitary landfill.
- ix) Project involving land clearing where 50% of the area or more having slopes exceeding 25 degrees (except quarry).
- x) Logging involving an area exceeding 500 hectares.

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- xi) Development of tourist or recreational facilities on islands in surrounding waters which are gazetted as national marine parks.
- xii) Construction of recovery plant (off-site) for lead-acid battery wastes.
- xiii) Scheduled wastes recovery or treatment facility generating significant amount of wastewater which is located upstream of public water supply intake.
- xiv) Non ferrous –primary smelting.

1.4 SelfRegulationofEnvironmentalManagement – RequirementofCompetentPerson (Amendmentof EQA in 2012)

Requirement of self-regulation on environmental management among industries in this country stipulated under Section 49A, EQA 1974 which require *Any owner or occupier of premise shall employ a person who has been certified by the Director General as a competent person to conduct all or any of the following activities:*

- i) The operation of a control equipment
- ii) The management of scheduled wastes
- iii) The conduct of studies
- iv) The preparation and submission of reports, plans, proposals, engineering drawings or other documents relating to environmental matters

Self-regulation approach which adopts preventive management and performance monitoring as a tool to monitor and evaluate the performance of the various unit operations and unit process of the pollution control systems is one the effective and the preferred approach in environmental pollution control practised by many environmental protection agencies worldwide.

The success of the operation of pollution control systems is primarily in the hands of the operators / supervisors. A Competent Person (knowledgeable and skill operator / supervisor) knows how to monitor the process occurring in his pollution equipment, understand the factors that affect its performance, the best operational conditions and how to maintain them, how to respond to upset conditions, understands the social obligations and legal implications for non-compliance.

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SOALAN/QUESTION :

1. What are the rules that are used to control of agro-based water pollution ?

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2. List prescribed activities under agricultures need to obtain EIA approval from the Director General of Environment

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3. What are the reason a project need to go through the Site Suitability Evaluation?

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4. What is regulation that are used to control industrial effluent?

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RUJUKAN / REFERENCES:

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